



Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
M. Tech. in Computer Science and Engineering
As per NEP 2020

S.Y.M. Tech. Computer Science and Engineering W.E.F. 2026-27

WIT
Dean (Academics)

Head
HEAD

Computer Sc. and Engg. Deptt.
W. I. T., Solapur

Computer Science and Engineering

Vision

To develop professional engineers in Computer Science & Engineering having ethical values, research aptitude and ability to address challenges of modernization in the IT industry aiming at overall sustainable development of the society.

Mission

- M1 - To impart quality education in the field of Computer Science & Engineering in accordance with the needs of the Modernization & Globalization through technology enabled education.
- M2 - To inculcate lifelong learning in students to face challenges posed by ever-changing IT career landscape as a disciplined professional with a sense of professional ethics.
- M3 - To inculcate critical thinking and creativity for identifying various societal issues and to provide solutions.
- M4 - To enhance career opportunities for students through academia-industry interaction and research.

Computer Science and Engineering

Post Graduate Programme

Program Educational Objectives (PEOs)

1. Graduates will exhibit the technical knowledge to provide innovative solutions for real-time problems.
2. Graduates will develop research aptitude and the lifelong learning to address the challenges in IT industries.
3. Graduates will pursue leadership skills, professionalism and ethical values during their career for the growth of society.

Program Outcomes (POs)

Postgraduates on successful completion of the programme will be able to:

PO1	Independently carryout research/investigation and development work to solve practical problems.
PO2	Write and present a substantial technical report/document.
PO3	Demonstrate a degree of mastery in Computer Science and Engineering; a level higher than the requirements in the appropriate bachelor program.
PO4	Analyze, evaluate and build systems in the field of Computer Science & Engineering.
PO5	Practice ethical behavior with professional code of conduct, effective managerial skills and life- long learning for solving real world problems related to social and environmental up liftment.
PO6	Acquire expertise in the latest trends in Computer Science & Engineering like data science, computer security and advanced computing leading to innovation.
PO7	Ability of life-long self learning and to keep one self up-to-date in the field of technology.

Legends used–

L	Lecture
T	Tutorial
P	Practical
D	Drawing
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
F.Y.	First Year
S.Y.	Second Year
M.Tech.	Master of Technology

Course Codes	
CC	Core Compulsory Course
SN*	Self-Learning <i>N*indicates the serial number of electives offered in the respective category</i>
EN*	Core Elective <i>N*indicates the serial number of electives offered in the respective category</i>
SM	Seminar
MP	Mini Project
PR	Project



WALCHAND INSTITUTE OF TECHNOLOGY
STRUCTURE OF M. Tech. (Computer Science and Engineering)
Choice Based Credit System Syllabus w.e.f. 2026-27
Structure (Scheme-24)

Semester III

Course Code	Name of the Course	Engagement Hours			Credits	FA	SA		Total
		L	T	P		ESE	ISE	ICA	
CMP3IK	Indian Science and Technology	2	-	-	2	-	50	-	50
CSP3IN	Internship	-	-	-	3	-	-	100	100
CSP3DS1	Dissertation Phase-I	-	-	30	15	-	200	150	350
Total		2	-	30	20	-	250	250	500

Guidelines for evaluation of Dissertation Phase I

Phase	Activity	Assessment Period	Marks
1	- Selection of dissertation topic - Presentation of dissertation topic in front of panel - Completion of dissertation synopsis and submission to the department	Within 30 days from the date of commencement of Semester III	100
2	Dissertation progress presentation	Last week of the second month of Semester III	100
3	Dissertation progress report submission and presentation	At the end of Semester III	150

Semester IV

Course Code	Name of the Course	Engagement Hours			Credits	FA	SA		Total
		L	T	P		ESE	ISE	ICA	
CMP4OE1N*	Open Elective (SWAYAM/NPTEL/ MOOC) #	-	-	-	3	-	-	-	-
CSP4DS2	Dissertation Phase-II	-	-	34	17	200	-	200	400
Total		-	-	34	20	200	-	200	400

Note:

For Open Elective course, students should enroll for one of the minimum 8 weeks duration courses offered by SWAYAM / NPTEL platform. They should complete its assignments and appear for certificate examination conducted by SWAYAM / NPTEL. Students should pass the examination till the end of Semester IV. Based on the marks obtained in the assignments and examination; credits will be transferred in Semester IV. The list of courses will be provided by the Board of Studies.

#As per the instruction provided in Semester II structure, the credits obtained by the students for the SWAYAM / NPTEL course will be transferred in this semester.

• **Dissertation Phase-II:**

ICA Progress seminar shall be delivered capturing details of the work done by student for dissertation at the end of Semester. Panel of Internal examiners will evaluate the Dissertation Phase-II.

NDY
 Dean (Academics)

are
 HEAD

Computer Sc. and Engg. Deptt.





WALCHAND INSTITUTE OF TECHNOLOGY
M. Tech. Part-II Sem-III (All Branches)
CMP3IK: INDIAN SCIENCE & TECHNOLOGY

Teaching Scheme:
Lecture: 2 hours/week, 2Credits

Examination scheme:
ISE: 50 Marks

COURSE OUTCOMES:

On completion of the course, the students will be:

1. Aware of the true history and rich culture of India.
2. Explain the scientific value of the traditional knowledge of India
3. Appreciate the knowledge of Metallurgy, Textile Chemistry & Pyro Technology in IKS
4. Elaborate Indian knowledge of Water Management and transportation systems
5. Describe the contribution of IKS in Mathematics & Astronomy.
6. Describe the concepts of IKS in Ecology and Environment.

1. Fundamentals (4 Hours)

- An overview of Indian contributions to technology
- Technological Innovations

2. Metallurgy, Textile Chemistry & Pyro Technology

(5 Hours)

- Copper/Bronze/Zinc: Important Mines (Zawar, Khetri mines)
- Iron and Wootz Steel Technology
- Textile and Dyeing-Indian Specialities (Kutchi Embroidery, Cotton Textile etc.)
- Ceramic Technology, Stone (Lapidary), Shell, Ivory, Faience & Glass Technology

3. Water Management & Transportation

(5Hours)

- Harappan and Traditional Water Management System of Gujarat
- Historical Sites-Sringeverpur, South Indian Water Management System, Western Ghats Cave-Kanheri, etc.
- Communities Involved in Water Management
- Modes of Transportations and Reforms
- Grand Trunk Road (Uttarapath & Dakshinapath)
- Development of Trading Techniques
- Boat & Ship Building

4. Mathematics & Astronomy

(5 Hours)

- Mathematics contained in the Sulbasutra
- Weaving Mathematics into Beautiful Poetry-Bhaskaracarya.
- The Evolution of Sine Function in India
- The Discovery of Calculus by Kerala Astronomers
- Vedanga Jyotish & Measuring Time & Calendar

5. Ecology and Environment

(5 Hours)

- Nakshatrara Gyaan and Agriculture
- Vernacular Architecture
- Forest Management and Urban Planning
- Agro forestry
- Tank, Lakes and Stepwells

6. India's Contribution in Science and Technology to the World

(4 Hours)

- Irrigation systems and practices in south India
- Physical structures of India
- Surgical Techniques
- Shipbuilding



Text Book:

- Introduction To Indian Knowledge System: Concepts And Applications Mahadevan, B. Bhat, Vinayak Rajat Nagendra Pavana R.N. Apr 2022 · Phi Learning Pvt. Ltd.

REFERENCES:

- R.M.Pujari, Pradeep Kolhe, N.R.Kumar, 'Pride of India: A Glimpse into India's Scientific Heritage', Samskrita Bharati Publication.
- 'Indian Contribution to Science', compiled by Vijnana Bharati.
- 'Knowledge traditions and practices of India', Kapil Kapoor, Michel Danino, CBSE, India.
- Dr. Subhash Kak, Computation in Ancient India, Mount, Meru Publishing (2016)
- Dharampal, Indian Science and Technology in the Eighteenth Century, Academy of Gandhian Studies, Hyderabad, 1971, republic. Other India Bookstore, Goa, 2000
- Robert Kanigel, The Man Who Knew Infinity: A Life of the Genius Ramanujan, Abacus, London, 1999
- Alok Kumar, Sciences of the Ancient Hindus: Unlocking Nature in the Pursuit of Salvation, Create Space Independent Publishing, 2014
- B.V. Subbarayappa, Science in India: A Historical Perspective, Rupa, New Delhi, 2013
- S. Balachandra Rao, Indian Mathematics and Astronomy: Some Landmarks, Jnana Deep Publications, Bangalore, 3rd edn., 2004
- S. Balachandra Rao, Vedic Mathematics and Science in Vedas, Navakarnataka Publications, Bengaluru, 2019
- Bibhutibhushan Datta, Ancient Hindu Geometry: The Science of the Śulba, 1932, repr. Cosmo Publications, New Delhi, 1993
- Bibhuti bhushan Datta & Avadhesh Narayan Singh, History of Hindu Mathematics, 1935, repr. Bharatiya Kala Prakashan, Delhi, 2004
- George Gheverghese Joseph, The Crest of the Peacock, Penguin Books, London & New Delhi, 2000
- J. McKim Malville & Lalit M. Gujral, Ancient Cities, Sacred Skies: Cosmic Geometries and City Planning in Ancient India, IGNCA & Aryan Books International, New Delhi, 2000.
- Clemency Montelle, Chasing Shadows: Mathematics, Astronomy and the Early History of Eclipse Reckoning, Johns Hopkins University Press, 2011
- Anisha Shekhar Mukherji, Jantar Mantar: Maharaja Sawai Jai Singh's Observatory in Delhi, AMBI Knowledge Resources, New Delhi, 2010
- Thanu Padmanabhan, (ed.), Astronomy in India: A Historical Perspective, Indian National Science Academy, New Delhi & Springer (India), 2010
- Acharya Prafulla Chandra Ray, A History of Hindu Chemistry, 1902, republ., Shaibya Prakashan Bibhag, centenary edition, Kolkata, 2002
- R. Balasubramaniam, Delhi Iron Pillar: New Insights, Indian Institute of Advance Study, Shimla & Aryan Books International, New Delhi, 2002
- R. Balasubramaniam, Marvels of Indian Iron through the Ages, Rupa & Infinity Foundation, New Delhi, 2008
- Anil Agarwal & Sunita Narain, (eds), Dying Wisdom: Rise, Fall and Potential of India's Traditional Water-Harvesting Systems, Centre for Science and Environment, New Delhi, 1997
- Fredrick W. Bunce: The Iconography of Water: Well and Tank Forms of the Indian Subcontinent, DK Printworld, New Delhi, 2013



WALCHAND INSTITUTE OF TECHNOLOGY
M.Tech. (Computer Science & Engineering) Part-II, Semester-III
CSP3DS1 – DISSERTATION PHASE –I

Teaching Scheme:
Practical : 30 Hours/week, 15 Credits

Examination Scheme
ISE–200Marks
ICA:150Marks

ISE(Synopsis Submission Seminar):

A student shall be expected to carry out intensive literature survey for a period of about two months in the field of interest and to select a topic for his/her dissertation in consultation with the faculty adviser assigned. The student shall then submit a report and deliver a seminar on the problem chosen by him/her to the panel of three departmental PG recognized faculty members. It shall be expected that a student justifies the gravity and also the relevance of the problem through his/her seminar. This shall be for the approval of synopsis.

Internal Continuous Assessment (ICA):

ICA based on the efforts put in by the student to carry out his/her work & the results obtained thereof.





WALCHAND INSTITUTE OF TECHNOLOGY
M.Tech. (Computer Science & Engineering) Part-II, Semester-III
CSP4OE1N* Open Elective (SWAYAM/NPTEL/MOOC) #

Engagement
Credits - 3

Examination Scheme

Note: The credits obtained by the student through a SWAYAM / NPTEL course will be transferred and considered under this semester.

Guidelines:

1. Course Selection:

Students should enrol for one of the minimum 8 weeks duration course offered by SWAYAM / NPTEL platform based on the list approved and provided by the BOS (Board of Studies) of the department.

2. Course Registration and Completion:

The student must register for the selected course and complete all associated assignments, quizzes, and activities as mandated by the platform (SWAYAM / NPTEL / MOOC).

3. Examination and Credit Transfer:

The student must register for the proctored final examination conducted by the respective platform.

Students should pass the examination till the end of Semester IV. Based on the marks obtained in the assignments and examination; credits will be transferred in Semester IV and submit the official result or scorecard to the department for credit transfer and academic record update.





WALCHAND INSTITUTE OF TECHNOLOGY
M.Tech. (Computer Science & Engineering) Part-II, Semester-IV
CSP4DS-DISSERTATION PHASE-II

Teaching Scheme:
Practical: 34 Hours/week, 17 Credits

Examination Scheme
ESE: 200 Marks
ICA: 200 Marks

Deliverables:

1. Recognize the importance of planning and preparation required to undertake a research project
 2. Extend the research work to a standardized level that can give solutions to real life problems.
 3. Demonstrate the ability to collate and critically assess/interpret data
 4. Develop an ability to effectively communicate knowledge in a scientific manner
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1. Students are expected to carry out independent research work on the chosen topic. In this semester it is expected that the student has carried out substantial research work including exhaustive literature survey, development/fabrication of experimental set-up (if any/required) and testing, and analysis of initial results thus obtained.
 2. It is expected that the student has completed most of the experimental/computation works and analysed the results so obtained as proposed in the synopsis.
 3. The work should be completed in all respects in this semester. The students are required to submit the dissertation work in the form of report as per the institute rule.
 4. The report shall be hard bound. It is a report on the work done by the student. It should have literature review, problem definition and formulation, adopted methodology, experimentation plan if any, results and discussion, conclusions, and its relevance to the further work.
 5. Student shall deliver all seminars using modern presentation tools. A hard copy of the report shall be submitted to the department before delivering the seminar. A PDF copy of the report must be submitted to the advisor along with other details if any.
 6. ICA Progress seminar shall be delivered capturing details of the work done by student for dissertation at the end of Semester. Panel of Internal examiners will evaluate the Dissertation Phase-II.
 7. The viva voce examination of the Dissertation-II shall consist of a presentation by the candidate and demonstration of the work carried out.

